AJAY THAKUR

Software Engineer | Full-Stack Developer | AI-Assisted Development

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Wollongong, NSW 2500 | Visa Subclass 485 — Full Work Rights in Australia

PROFILE

Master of Computer Science graduate (Software Engineering) from the University of Wollongong with hands-on experience across full-stack web development, cloud-integrated systems, and AI API integration. Brings a strong engineering foundation, real team leadership experience, and the ability to work independently to ship production-ready projects. Comfortable working across the stack and with modern AI tools to build and iterate faster.

TECHNICAL SKILLS

Languages & Engines	C++, C# (Unity), Python, PHP, HTML/CSS, SQL, JavaScript
Frameworks & APIs	Flask, REST APIs, Unity (2D game design, gameplay mechanics)
Cloud & Tools	Microsoft Azure, Vercel, Git/GitHub, VS Code, Microsoft 365
Databases	SQL, SQLite, Cloud-hosted databases
AI & Prompting	Groq API, LLM integration, Prompt engineering, AI-assisted development
Software Practices	Data Structures & Algorithms, Unit & Integration Testing, CI/CD, Agile workflows

EDUCATION

Master of Computer Science — Major: Software Engineering *2022 – 2024*

University of Wollongong, NSW, Australia

- Capstone Project: QR Code-Based Attendance System — Python, Flask, SQL, PHP, Azure
- Designed secure real-time data handling and cloud integration for attendance tracking at scale

Bachelor of Computer Science *2015 – 2018*

Government Degree College Kullu, India

- Final Project: 2D Game Development in Unity (C#) — designed gameplay mechanics and interactive systems

PROJECTS

AI Chat Assistant

- Built a fully functional AI-powered chat assistant using HTML, CSS, and JavaScript integrated with the Groq API (Llama 3.3) for real-time AI responses
- Implemented a secure backend proxy using Vercel serverless functions to protect the API key — key is never exposed in source code or the browser
- Frontend deployed on GitHub Pages, all API calls routed securely through a Vercel server demonstrating a real-world production architecture

- Features: chat history via localStorage, typing indicators, markdown formatting, code block rendering, mobile-responsive sidebar

Tech: HTML/CSS, JavaScript, Groq API, Llama 3.3, REST APIs, localStorage, Vercel, Serverless Functions, GitHub Pages

QR Code-Based Attendance System (Capstone)

- Architected a backend service with Python and Flask to process real-time QR scan events
- Implemented secure data storage and seamless API integration with a cloud-hosted SQL database
- Deployed on Microsoft Azure with focus on reliability and data integrity at scale

Tech: Python, Flask, SQL, Azure, REST APIs, PHP

2D Game Development in Unity

- Designed and implemented a complete 2D interactive game from scratch using Unity
- Built gameplay loop, physics interactions, and level design with a focus on player engagement
- Applied OOP patterns in C# to structure game systems and entity logic

Tech: C#, Unity, OOP, Game Design, Physics Systems

WORK EXPERIENCE

Retail Sales & IT Support | **Cignall** *Aug 2024 – Present*

- Provided IT and digital support for internal systems, troubleshooting hardware/software issues in a fast-paced environment
- Led a team of 20 employees, managing schedules and coordinating daily operations
- Identified patterns in data and adjusted strategies to improve service delivery, developing an analytical mindset

Service Team Member | **Coles** *Apr 2022 – Present*

- Collaborated within a cross-functional team to deliver excellent customer outcomes under high-volume, time-pressured conditions
- Demonstrated reliability and adaptability across shifting priorities in a large-scale retail operation

Office & IT Supervisor | **Zenith Event & Services** *Dec 2018 – Dec 2021*

- Supervised IT infrastructure and office systems for 50 employees, managing hardware, software, and data operations
- Trained staff in office software and operational procedures, developing strong communication and documentation skills

CERTIFICATIONS

- **Web Designing & Development Certificate** — 2015/2016
- **Workshop on Online Safety | Social Surfing 3.0**

References available upon request